NISAR AHMAD

Computer Science Undergraduate | Networks & Operating Systems · Cloud Computing · Distributed Systems
iamnisarahmad08@gmail.com | Punjab, Pakistan | linkedin.com/in/nisar-ahmad-973a673a2 | github.com/nisar-ai

PROFILE

Third-semester BS Computer Science student (GPA 3.50/4.00; top five in a class of 50) at COMSATS University Islamabad, Sahiwal Campus, with a research focus on computer systems and cloud computing. I build and measure systems software: a discrete-event microservice simulator for studying queueing, retries and tail latency, and a serverless token-bucket rate limiter. My work emphasises reproducible experiments, performance evaluation and clean implementation. Seeking a summer 2027 research internship in cloud computing and computer systems at EPFL.

EDUCATION

COMSATS University Islamabad, Sahiwal Campus | BS Computer Science **Sep 2025 – Jul 2029 | GPA 3.50/4.00**
Completed: Programming Fundamentals; Object Oriented Programming (Java); Discrete Structures
In progress (Fall 2026): Data Structures (C++); Database Systems (Oracle SQL); Fundamentals of Digital Logic Design
Planned (Spring 2027): Computer Networks; Artificial Intelligence; Information Security

RESEARCH & SYSTEMS PROJECTS

CloudTrace Lab — Discrete-Event Microservice Simulator github.com/nisar-ai/cloudtrace-lab
Python · Gradio · NumPy · Pandas · Plotly · Hugging Face Spaces

- Designed and implemented a configurable discrete-event simulator of a microservice pipeline, modelling request arrivals, finite service capacity, queueing, simulated failures, and retries with exponential backoff and jitter.
- Built seeded, reproducible experiments with request- and service-level CSV trace export, enabling analysis of mean and p50/p95/p99 latency, success rate, utilisation, queue wait and bottlenecks.
- Example run (300 requests): 298 succeeded (99.3%); p50 3.32 s, p95 6.05 s, p99 6.26 s (simulated time); 24 requests needed more than one attempt.

Token Bucket Rate Limiter with Live Traffic Simulator
JavaScript · Netlify Functions · Serverless · REST APIs

- Implemented a serverless token-bucket rate-limiting API (token refill, burst handling, HTTP 429 responses) on Netlify Functions.
- Built a live traffic simulator with throttling visualisation and timestamped event logs; deployed through Git-based continuous deployment.

Other Projects

- **UniBot Next** (Next.js, React, Tailwind CSS, Groq API, Netlify): AI admissions chatbot with server-side API handling and environment-variable secret management.
- **GlobalVisa Flow** (Next.js, TypeScript; in development): multilingual visa-planning app with document checklists and cost/timeline estimates; tested locally.
- **Multilingual Translator** (Streamlit, NLLB-200, Hugging Face Spaces) · **NovaCalc** scientific calculator with a controlled expression evaluator · **Weather Forecast App** using public weather APIs.

EXPERIENCE

Independent Software & Cloud Systems Developer **Feb 2026 – Present**

- Developed and deployed applications on Netlify and Hugging Face Spaces, spanning frontends, serverless backends and third-party API integration.
- Studied queueing, rate limiting, retry strategies and tail-latency analysis through implementation and simulation.
- Used Git/GitHub for version control; reviewed and validated AI-assisted code against expected behaviour.

TECHNICAL SKILLS

Programming: Python, Java, C++, JavaScript, TypeScript, SQL (Oracle), HTML5, CSS3
Systems & Cloud (applied in projects): computer systems, cloud computing, distributed-systems concepts, microservices, discrete-event simulation, queueing and capacity modelling, rate limiting, exponential backoff and jitter, tail-latency analysis (p50/p95/p99), serverless architecture, REST API design, performance analysis
Frameworks & Tools: Node.js, Next.js, React, Tailwind CSS, Gradio, Streamlit, NumPy, Pandas, Plotly, PyTorch, Hugging Face Transformers, Git/GitHub
Deployment: Netlify, Netlify Functions, Hugging Face Spaces, Git-based continuous deployment

ACHIEVEMENTS & CERTIFICATIONS

- **Awardee, Chief Minister's Laptop Scheme.**
- **Top-five performer in a class of 50 students** (COMSATS University Islamabad, Sahiwal Campus).
- **Generative & Agentic AI Application Developer Certificate**, Pak Angels, Cohort 11 (Aug–Oct 2026) · **Generative AI Application Developer Certificate**, Cohorts 2 and 3.
- **Aspire Leadership Programme** (enrolled, Oct 21 – Dec 1, 2026) · **Responsive Web Design**, freeCodeCamp.

REFERENCE

Dr. Muhammad Shoaib, Lecturer, Department of Computer Science
COMSATS University Islamabad (CUI), Sahiwal Campus, Pakistan | Email: mshoaib@cuisahiwal.edu.pk